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Three-Dimensional Compressible Chemically Reacting Computational Fluid Dynamics with Tensor Trains

Sinan Demir*

Argonne National Laboratory, Lemont, IL 60439, USA

Ryan F. Johnson[†], Brian T. Bojko[‡], Andrew Corrigan[§], David A. Kessler[¶]

Laboratories for Computational Physics and Fluid Dynamics, Naval Research Laboratory, Washington, DC 20375-5344, USA

Pierson Guthrey^{||}, Jason Burmark^{**}

Lawrence Livermore National Laboratory, Livermore, CA 94550, USA

Samuel Irving^{††}

Formerly Naval Information Warfare Center (NIWC) Atlantic

This study assesses the accuracy and efficiency of a framework employing tensor trains to represent flamelet progress variable manifolds in chemically reacting computational fluid dynamics (CFD). The CFD with tensor train assistance is compared to an established chemically reacting CFD framework: a stable explicit time integration scheme with direct source term application via the flamelet progress variable approach (FPV). An overview of the FPV approach is given with the objective of isolating where the tensor trains can be employed. The feasibility of storing flamelet-generated manifolds (FGM) in the tensor train format is explored, encompassing each species considered for both a simple 13-species hydrocarbon model and a 95-species UCS mechanism. The study sheds light on how tensor trains can be seamlessly integrated into existing CFD frameworks, emphasizing the potential of tensor networks for advancing CFD technologies, particularly in high-order numerics. A baseline backward-facing step case in 2D and 3D is utilized to demonstrate the interchangeability of the tensor network with the FPV. Noteworthy compressibility features, such as table memory reduction, are discussed, along with other potential efficiencies associated with these promising approaches. The combustion dynamics were simulated and incorporated into an existing discontinuous Galerkin finite element solver provided by the JENRE Multiphysics Framework.

I. Introduction

In computational fluid dynamics (CFD), dimensionality reduction techniques are important for efficiently managing high-dimensional data in complex fluid flow simulations. These methods streamline computations, lower memory requirements, and aid in analyzing and interpreting simulation results. One noteworthy approach, Proper Orthogonal Decomposition (POD) [1], excels in identifying dominant spatial modes of variation in the flow field. These selectively retained POD modes play a crucial role in capturing significant structures and variations, thereby contributing significantly to computational efficiency. POD has found successful applications in various CFD domains, such as aerodynamics [2], heat transfer [3], and combustion [4, 5].

Dynamic Mode Decomposition (DMD) complements POD by extracting spatiotemporal coherent structures, aiding in a thorough comprehension of their temporal behavior. DMD finds applications in studying diverse fluid flow

*Research Scientist, Computational Multi-Physics, AIAA Member, sdemir@anl.gov

[†]Aerospace Engineer, Laboratory for Propulsion, Energetics, and Dynamic Systems, Code 6041, AIAA Member, ryan.johnson@nrl.navy.mil

[‡]Aerospace Engineer, Laboratory for Propulsion, Energetics, and Dynamic Systems, Code 6041, AIAA Member.

[§]Research Mathematician, Laboratory for Propulsion, Energetics, and Dynamic Systems, Code 6041, AIAA Member.

[¶]Mechanical Engineer, Laboratory for Multiscale Reactive Flow Physics, Code 6043, AIAA Member.

^{||}Code Physicist, Weapon Simulation and Computing, guthrey1@llnl.gov

^{**}Applications, Simulations, and Quality, burmark1@llnl.gov

^{††}Computer Scientist

phenomena, including shock-turbulent boundary layer interactions [6], transition to turbulence [7], flow control [8] and combustion [9]. Karhunen-Loève Expansion (KLE) is an additional mathematical technique [10] that effectively reduces the dimensionality of turbulent flow data by representing a turbulent flow field through an expansion of orthogonal modes.

Nonlinear dimensionality reduction (NLDR) techniques, such as t-distributed Stochastic Neighbor Embedding (t-SNE) and Isomap, are particularly useful for handling complex flow scenarios with nonlinear relationships. NLDR techniques can be used to visualize high-dimensional CFD data and identify patterns that would be difficult to discern using traditional linear methods [11].

Machine learning techniques [12], including autoencoders [13] and neural networks [14], constitute a powerful arsenal for data-driven dimensionality reduction in CFD, learning compact representations of the flow field for efficient encoding and decoding. Simultaneously, Reduced Order Modeling (ROM) leverages techniques like POD to create low-dimensional representations applicable for real-time simulations and optimization, maintaining a balance between accuracy and computational efficiency [15, 16]. This diverse spectrum of dimensionality reduction techniques in CFD caters specifically to flow characteristics and simulation objectives, enhancing computational efficiency and providing valuable insights into the complexities of fluid flow phenomena.

In the context of turbulent combustion simulations, reduced-order manifold-based flamelet approaches have emerged as a highly constructive alternative to detailed finite rate chemistry (FRC) models. The FRC model with detailed chemical mechanisms provides a high-fidelity representation of the combustion process, allowing for accurate predictions of species concentrations, temperature profiles, and reaction rates. It comprehensively models the transport of individual chemical species in a reacting flow, encompassing considerations for advection, diffusion, and chemical reactions of each species within the surrounding fluid. The chemical mechanisms detailing the chemical reactions are numerically solved to determine the rates at which different species are formed and consumed over time. This typically involves solving a stiff system of ordinary differential equations (ODEs) that represent the dynamic evolution of species concentrations during the combustion process.

However, due to the large number of species and reactions, the computational intensity of this approach can be challenging, necessitating advanced numerical methods and significant computational resources. In addition to calculating reaction rates, another challenge is that the source terms can be stiff due to the substantial difference in time scales. The time scales associated with different chemical reactions can differ by orders of magnitude, and the source terms are much faster than transport time scales.

A particular promising family of approaches, combined referred to as the flamelet progress variable (FPV) approach, has been used to reduce the chemically reacting compressible CFD state to several transported scalars. Instead of directly simulating the detailed chemical kinetics of combustion, flamelet manifolds are precomputed and tabulated. Each flamelet corresponds to a specific state of combustion, and the manifold captures the local flame structure, temperature, and associated species concentrations. During CFD simulations, thermochemical quantities are interpolated from these tables. Our collaborative group has successfully had success using the unsteady flamelet progress variable UFPV [17–20] from Argonne National Laboratory (ANL), as well as FPV and the FPV-DynaSTE [21, 22] approach from the Naval Research Laboratory (NRL).

Flamelet-based manifold reduction techniques are essential for handling the computational complexity inherent in the high-dimensional nature of combustion. By reducing the manifold size, these techniques enhance the efficient handling of detailed chemical kinetics, enabling faster simulations in turbulent combustion and propulsion systems. Manifold reduction not only addresses computational efficiency but also plays a crucial role in managing numerical instability, making these techniques well-suited for large-scale, highly-accurate CFD simulations of turbulent combustion.

The reliance on lower-dimensional manifold assumptions often hinders the flamelet model’s ability to capture intricate turbulent combustion physics precisely. Addressing this limitation necessitates using higher-dimensional flamelet manifolds, mainly when predicting phenomena such as compression/expansion waves and heat transfer effects through walls. Incorporating additional variables, like pressure, into flamelet tables is essential for accurate predictions. However, expanding the dimensions of these tables introduces challenges, notably high-dimensional interpolation of parameterized variables, leading to increased memory allocation and computational time. This phenomenon, known as the curse of dimensionality, results in non-linear growth of computational requirements, making the model less efficient than the detailed kinetics it seeks to replace. In essence, the expansion of flamelet model dimensions amplifies both memory footprint and retrieval costs, impacting the overall efficiency of the model.

Our group and collaborators have successfully employed strategies using artificial neural networks (ANNs) to overcome the challenges of high memory footprint and retrieval costs associated with large flamelet tables [17, 23, 24]. The primary objective of the ANN-based tabulation is to establish a relationship between flamelet manifold variables and

target outputs, such as species mass fractions, temperature, and progress variable source terms. A Grouped Multi-Target ANN (GMT-ANN) framework [23] groups strongly correlated species, using a unique ANN for each group to map inputs to species' mass fractions. A more recent approach, the Mixture of Experts (MoE) [24], autonomously bifurcates flamelet manifolds by assigning regression tasks among specialized ANNs. Each neural network specializes in a specific input space portion, determined by a gating network—a neural network classifier. MoE facilitates automatic bifurcation using non-linear planes in high-dimensional turbulent combustion manifolds, addressing complex behavior in different zones.

GMT-ANN and MoE-based flamelet manifold representations have demonstrated significant memory and retrieval cost savings while maintaining high accuracy in both a priori and a posteriori studies, particularly in Large Eddy Simulations (LES) of hydrocarbon fuel combustion in spray flames and engines [23, 24]. We are currently implementing ANNs into an existing discontinuous Galerkin solver provided by the JENRE Multiphysics Framework, alongside the UFPV [17, 18, 20] approach, with a focus on large-scale simulations of high-speed combustion in complex configurations.

When employing ANNs to represent the flamelet manifold, training the progress variable source terms becomes notably challenging. This difficulty arises from the fact that these values can undergo significant changes across multiple orders of magnitude throughout the reaction process. Even a small error in the network can lead to a substantial magnitude error during the inference stage. Consequently, more effort is required to accurately capture the source terms during neural network training, as discussed in detail in [18]. Our motivation for this work stems from a collective interest in extending our focus beyond ANNs to efficiently compress and represent the FPV, particularly through the use of tensor trains.

We present here the feasibility of utilizing tensor trains to store FPV tables. Tensor networks [25], known for their effectiveness in compressing multidimensional data, are considered for creating estimations of FPV tables. The memory requirements of tensor trains are shown to be favorable compared to conventional lookup tables, especially when the ranks of the tensor train are small. The compression rate, defined as the ratio of memory requirements between the full table and the tensor train, is introduced as a quantification of this benefit. The goal is to choose a tolerance for each tabulated species, ensuring that the tensor train estimation maintains an acceptable error. Future work will involve more precise estimates, and a table is presented to showcase compression rates and errors for an exemplary species. This study provides a constructive demonstration of the feasibility of employing tensor trains for progress variable tables in CFD applications. We comment on compression rates using tensor trains for both a simple 13-species hydrocarbon model and a 95-species UCS mechanism [26]. A baseline backward-facing step case in 2D and 3D is utilized to demonstrate the interchangeability of the tensor network with an FPV table with bilinear interpolation.

II. Governing equations

Here we present the compressible, multicomponent, chemically reacting Navier-Stokes equations to demonstrate the reduction of size achieved by the FPV approach. The Navier-Stokes equations, in d spatial dimensions, are given by

$$\frac{\partial y}{\partial t} + \nabla \cdot \mathcal{F}(y, \nabla y) - S(y) = 0 \quad (\text{II.1})$$

where y is the state vector, ∇y is its spatial gradient, t is time, $\mathcal{F}$ is the flux, and $S = (0, \dots, 0, \omega_1, \dots, \omega_{n_s})^T$ is the chemical source term, with ω_i corresponding to the production rate of the i th species. The physical coordinates are denoted by $x = (x_1, \dots, x_d)$. The vector of state variables is expanded as

$$y = (\rho v_1, \dots, \rho v_d, \rho E, C_1, \dots, C_{n_s})^T, \quad (\text{II.2})$$

where ρ is density, $v = (v_1, \dots, v_d)$ is the velocity vector, E is the specific total energy, $C = (C_1, \dots, C_{n_s})$ is the vector of molar concentrations, and n_s is the number of species. The partial density of the i th species is defined as

$$\rho_i = W_i C_i,$$

where W_i is the molecular weight of the i th species, from which the density can be computed as

$$\rho = \sum_{i=1}^{n_s} \rho_i.$$

The mole and mass fractions of the i th species are given by

$$X_i = \frac{C_i}{\sum_{i=1}^{n_s} C_i}, \quad Y_i = \frac{\rho_i}{\rho}.$$

The equation of state for the mixture is written as

$$P = R^0 T \sum_{i=1}^{n_s} C_i, \quad (II.3)$$

where P is the pressure, T is the temperature, and R^0 is the universal gas constant. The specific total energy is the sum of the mixture-averaged specific internal energy, u , and the specific kinetic energy, written as

$$E = u + \frac{1}{2} \sum_{k=1}^d v_k v_k,$$

where the former is the mass-weighted sum of the specific internal energies of each species, given by

$$u = \sum_{i=1}^{n_s} Y_i u_i.$$

With the thermally perfect gas model, u_i is defined as

$$u_i = h_i - R_i T = h_{\text{ref},i} + \int_{T_{\text{ref}}}^T c_{p,i}(\tau) d\tau - R_i T,$$

where h_i is the specific enthalpy of the i th species, $R_i = R^0/W_i$, T_{ref} is the reference temperature of 298.15 K, $h_{\text{ref},i}$ is the reference-state species formation enthalpy, and $c_{p,i}$ is the specific heat at constant pressure of the i th species, which is approximated with a polynomial as a function of temperature based on the NASA coefficients [27, 28], i.e.,

$$c_{p,i} = \sum_{k=0}^{n_p} a_{ik} T^k. \quad (II.4)$$

The flux can be expressed as the difference between the convective flux, $\mathcal{F}^c$, and the viscous flux, $\mathcal{F}^v$, i.e.,

$$\mathcal{F}(y, \nabla y) = (\mathcal{F}^c(y) - \mathcal{F}^v(y, \nabla y)),$$

where the k th spatial components are defined as

$$\mathcal{F}_k^c(y) = (\rho v_k v_1 + P \delta_{k1}, \dots, \rho v_k v_d + P \delta_{kd}, v_k(\rho E + P), v_k C_1, \dots, v_k C_{n_s})^T \quad (II.5)$$

and

$$\mathcal{F}_k^v(y, \nabla y) = \left(\tau_{1k}, \dots, \tau_{dk}, \sum_{j=1}^d \tau_{kj} v_j + \sum_{i=1}^{n_s} W_i C_i h_i V_{ik} - q_k, C_1 V_{1k}, \dots, C_{n_s} V_{n_s k} \right)^T, \quad (II.6)$$

respectively. τ is the viscous stress tensor, q is the heat flux, and V_{ik} is the k th spatial component.

In the context of the FPV approach, the assumption can be simplified to utilize only a limited number of progress variables instead of the detailed large number of species required for finite rate kinetics [17, 18, 21, 22].

To develop the FPV approach for solving the reacting Navier-Stokes equations, the first step involves creating Flamelet-Generated Manifolds (FGMs) and storing them in the flamelet table. These FGMs are essential for accessing species, thermodynamic properties, and source terms during runtime, encompassing all relevant combustion information for the studied phenomena. The FGMs play a crucial role in providing the necessary data for the FPV approach and are instrumental in capturing the combustion characteristics in the simulations. The FPV methodology utilizes a mixture fraction and progress variable. The mixture fraction a conserved scalar defined by a linear summation of species mass fractions and is express below as,

$$Z_C = \sum_{i=1}^n \frac{a_{iC} MW_C}{MW_i} Y_i \quad (\text{II.7})$$

where, Z_C is the mixture fraction as defined for all species containing carbon atoms, a_{iC} is the number of carbon atoms within a molecule - e.g. for CH_4 $a_{iC} = 1$ and for C_2H_2 $a_{iC} = 2$, MW is the molecular weight of carbon with subscript C or of the i^{th} species with subscript i , and Y_i is the mass fraction. The definition of the mixture fraction illustrates its conserved scalar property by representing the mass fraction of the fuel stream in the mixture. It signifies the distributed carbon atoms within a system, ensuring the conservation of atoms. The mixture fraction can be defined with any molecule contained in the fuel species, as it varies from $Z = 1$ at the fuel to $Z = 0$ at the oxidizer. Notably, it maintains the same distribution if Z_H is used, provided the diffusivities of species are assumed to be equal.

The definition of the mixture fraction tracks the mixing of fuel and oxidizer, while an additional term is required to define the extent of reaction and the variation in flame shape due to varying strain rates. This is achieved through the reaction progress variable term as shown below, which accounts for the evolution of the combustion process and the associated changes in flame shape.

$$C_{pv} = \frac{\sum \alpha_i Y_i}{\sum \alpha_i (Y_i, eq)_{st}} \quad (\text{II.8})$$

$$\Lambda_C = C_{pv, st} = \frac{\sum \alpha_i Y_{i, st}}{\sum \alpha_i (Y_i, eq)_{st}} \quad (\text{II.9})$$

here, C_{pv} is the progress variable and α_i are the weights used to scale the mass fractions. This value is normalized between 0 and 1 using the Y_i 's at the stoichiometric - subscript (st) - limit and at the equilibrium solution (subscript (eq)). Λ_C is the value of C_{pv} at the stoichiometric condition and can be used to catalog all flame data as a function of Z using a singular value. This parameter is used to optimize the weights α_i during the FGM development process. It is also important to note, the C_{pv} term is not a conserved scalar and will therefore have a source term, $\dot{S}_{C_{pv}} = \sum \alpha_i \dot{\omega}_i / \sum \alpha_j (Y_{j, eq})_{st}$, that needs to be tabulated and looked up during runtime when solving the set of partial differential equations describing the FPV approach. With definitions Z and C_{pv} , it is now possible to tabulate all necessary species data and source terms as a function of these two parameters. Further details on the tabulated methodology and optimization strategies can be found in the following references [29–31].

Given the definitions of mixture fraction and progress variable, as provided in Eqs.(II.7) and (II.8) respectively, it is possible to modify the species transport equations to readily obtain transport equations for Z and C_{pv} . Assuming constant diffusivity allows for the definition of Z as defined in Eq.(II.7), to be used to create a linear summation of all species equations to obtain the transport equation for the mixture fraction. This results in a cancellation of the source terms $\dot{\omega}_i$ because any creation or destruction of species containing elemental carbon will balance, leading to the conserved scalar transport equation. Similarly, the definition of C_{pv} can be created by multiplying and dividing through by the weights and constant values at the equilibrium solution, after summation, the transport equation for the progress variable is obtained. In this instance, C_{pv} is not a conserved scalar and will contain the species weighted source term $\dot{S}_{C_{pv}}$ to describe the creation and destruction of the reaction progress variable in relation to the 1D flame solution. Thus the new state, convective, and diffusive terms used to solve combustion problems using FPV within the JENRE Multiphysics Framework are as follows:

$$y = (\rho, \rho Z, \rho C_{pv}, \rho v_1, \dots, \rho v_n, \rho E)^T \quad (\text{II.10})$$

$$\mathcal{F}_k^c(y) = (\rho v_k, \rho Z v_k, \rho C_{pv} v_k, \rho v_k v_1 + p I, \dots, \rho v_k v_1 + p I, (\rho E + p) v_k)^T \quad (\text{II.11})$$

$$\mathcal{F}_k^v(y) = \left(0, \rho D Z, \rho D C_{pv}, \tau_{1k}, \dots, \tau_{dk}, \lambda \nabla T + \sum_{j=1}^d \tau_{kj} v_j - \sum_i^{n_s} \rho D h_i \frac{\partial Y_i}{\partial x_k} \right)^T \quad (\text{II.12})$$

$$\mathcal{S}(y) = (0, 0, \dot{S}_{C_{pv}}, 0, \dots, 0, 0)^T \quad (\text{II.13})$$

In large chemically reacting systems, the FPV state (Eq. II.10) involves fewer equations compared to the FRC state (II.2). Specifically, when employing the FPV with the GRI methane combustion mechanism [32], the system is reduced from 52 species to three scalars: density, mixture fraction, and progress variable. This reduction not only lowers the cost of evaluating numerous chemical reactions but also has the potential to eliminate stiffness, thereby avoiding the need for expensive implicit algorithms that are challenging to solve, load balance, and scale.

It is worth noting that in the FPV approach, additional scalars, such as mixture fraction variance, may be used when applying an explicit subgrid scale model. However, in this work, we focus solely on the basic FPV formulation. Despite the assumptions required to reduce the order of the problem, the FPV approach can still capture rich kinetic information, making it an appealing formulation for modeling. However, as discussed in the next section, the manifolds can become cumbersome and serve as the main limiting factor when transitioning FPV models to account for larger-scale fuels and cover a broader parameter space.

III. Flamelet Progress Variable Manifolds

One-dimensional flames are systematically computed across a range of conditions relevant to the target simulation. This involves setting boundary conditions identical to those in the large-scale simulation, utilizing the same fuel and oxidizer compositions at similar temperatures and pressures. The data of these flame solutions are stored in a manifold. At any point in the simulation the full chemical state of a flame can be recovered under the limitations of constructed table by extracting (Z, C_{pv}) from the state y . A generic approach employing the compressible, total energy formulation outlined earlier can be described as follows:

- 1) Advance $\frac{\partial y}{\partial t} + \nabla \cdot \mathcal{F}(y, \nabla y) - \mathcal{S}(y) = 0$ in time using spatial and temporal discretization from [33].
- 2) Extract the FPV manifold independent variables from the state, using ρ , ρZ , and ρC_{pv} from y
- 3) Use the lookup table to estimate the species mass fraction and progress variable source term, $(Z, C_{pv}) \xrightarrow{\text{lookup table}} (Y_1, \dots, Y_{n_s}, \dot{S}_{C_{pv}}''')$
- 4) Use ρ , from y to estimate the concentrations $(C_1, \dots, C_{n_s})$ using $(Y_1, \dots, Y_{n_s})$
- 5) Perform a Newton solve for T from ρE using $(C_1, \dots, C_{n_s})$
- 6) Use T in the equation of state, II.3, to estimate pressure, P .
- 7) Apply the source term $\dot{S}_{C_{pv}}'''$ and pressure, P , to advance the solution via a return to 1).

It is important to note that a variety of methodologies exist in the family of FPV approaches that we have employed. For instance, we have seen that it is advantageous to contract the state gradient with a temperature jacobian, $\nabla T = \nabla y T'$, rather than evaluating the temperature gradient, ∇T , where the temperature jacobian, T' , may rely on jacobians of the species that can be calculated via the flamelet manifold. Additionally the manifold could grow in independent variables to cover more variability in physics, for instance $(Z, C_{pv}, Z') \xrightarrow{\text{lookup table}} (Y_1, \dots, Y_{n_s}, \dot{S}_{C_{pv}}''')$ where Z' is mixture fraction

variance or even $(Z, C_{pv}, Z', \chi, P_{prev}) \xrightarrow{\text{lookup table}} (Y_1, \dots, Y_{n_s}, \dot{S}_{C_{pv}}''')$ where P_{prev} is the pressure at the previous solution

and χ are scalar dissipation rates (such as with the UFPV approach [17, 18]). The $\xrightarrow{\text{lookup table}}$ process is fairly dependent on the manifold's incorporation into a CFD code and the underlying physics modeled. In the results presented here, we use a Bisection search to find the nearest index for Z and C_{pv} and, using those indices, we perform a bilinear interpolation.

The size of these tables scales with increasing complexity. If we assume minimal overhead to array storage, the size in memory of the manifolds we consider is:

$$\text{Memory Usage} = n_s \times n_Z \times n_{C_{pv}} \times d \quad (\text{III.1})$$

Where d represents the necessary memory footprint to store the desired scalar, with double-precision taking up 8 bytes in this case. Additionally, n_Z and $n_{C_{pv}}$ denote the degrees of freedom for which flame solutions are generated. In our observations, n_Z and $n_{C_{pv}}$ typically range from approximately 100 to rarely exceeding 500.

This leads to a conservative estimate of $95 \times 500 \times 500 \times 8 \approx 192$ MB for the USC mechanism. While this may not seem substantial, it is important to note that a slight increase in dimensionality can quickly deplete hardware memory. If the objective is to enhance the manifold's dimensions to incorporate higher fidelity, such as including scalar dissipation χ , P_{prev} , with dimensionality increasing by $n_\chi \approx 100$ and $n_{P_{prev}} \approx 100$ degrees of freedom, along with the use of more complex fuels like a jet-fuel surrogate containing $n_s = 362$ species [34], the memory usage becomes:

$$\text{Memory Usage} = n_s \times n_Z \times n_{C_{pv}} \times n_\chi \times n_{P_{prev}} \times d, \quad (\text{III.2})$$

which evaluates to $362 \times 500 \times 500 \times 100 \times 100 \times 8 \approx 724$ GB. Even on the most advanced hardware, achieving the necessary memory space that is shared with the essential CFD algorithms would most likely necessitate a two-orders-of-magnitude compression.

IV. Tensor Trains for FPV Tables

Tensor networks have been exploding in popularity due to their ability to compress multidimensional representations of data [35–37], accelerate neural networks [38, 39], and solve challenging PDE systems [40–42].

In this work we consider the feasibility of storing the FPV tables in the tensor train format. For example, given a pointwise defined function f in three dimensions, we are using tensor trains to create an estimation F ,

$$f_{ijk} \approx F_{ijk} = \sum_{r_1}^{R_1} C_{r_0 i r_1}^{(0)} \otimes \sum_{r_2}^{R_2} C_{r_1 j r_2}^{(1)} \otimes C_{r_2 k r_3}^{(2)}, \quad (\text{IV.1})$$

where N_i are the table extents, ijk are the table indices $i < N_0, j < N_1, k < N_2$, $r_i < R_i$ for $i = 0, 1, 2, 3$ with $R_0 = R_3 = 1$ are the tensor train ranks, and $C_{\ell mn}^{(d)}$ for $d = 0, 1, 2$ are third order tensors known as the tensor train cores. The ranks R_i depend on a mix of the structure of the underlying data and a specified tolerance, see [35]. The tensor train memory requirements are computable as

$$\text{memory, tensor train} = N_0 R_1 + N_1 R_1 R_2 + N_2 R_2, \quad (\text{IV.2})$$

so if the ranks R_i are small, the memory requirements of the tensor train table are much more favorable than the conventional lookup table. We quantify this benefit as the compression rate, defined as

$$\text{compression rate} = \frac{\text{memory, full table}}{\text{memory, tensor train}} = \frac{N_0 N_1 N_2}{N_0 R_1 + N_1 R_1 R_2 + N_2 R_2}. \quad (\text{IV.3})$$

It is also important to quantify the estimation error introduced by the tensor train, which we simply compute as the relative error

$$\text{estimation error} = \frac{\|f - F\|_\infty}{\|f\|_\infty}. \quad (\text{IV.4})$$

The goal of this study is to choose a tolerance for each tabulated specie such that the tensor train estimation for that species has an acceptable error. In the context of the FPV approach, the table values are interpolated in a multi-linear fashion, which also introduces error. Thus we can rigorously choose an acceptable error by requiring that it is less than than the error introduced by interpolation. In this work we use a simple estimate defined by

$$\text{interpolation error} = \max_{i < N_0, j < N_1, k < N_2} \left(f_{ijk}^{\max} - f_{ijk}^{\min} \right), \quad (\text{IV.5})$$

where

$$f_{ijk}^{\max} = \max_{\hat{i} \in \{i, i+1\}, \hat{j} \in \{j, j+1\}, \hat{k} \in \{k, k+1\}} f_{\hat{i} \hat{j} \hat{k}}, \quad (\text{IV.6})$$

and

$$f_{ijk}^{\min} = \min_{\hat{i} \in \{i, i+1\}, \hat{j} \in \{j, j+1\}, \hat{k} \in \{k, k+1\}} f_{\hat{i} \hat{j} \hat{k}}. \quad (\text{IV.7})$$

In future work we will consider sharper estimates.

In Table 1, we show various compression rates and errors as a function of the tensor train tolerance for an exemplary specie, CH_4 , for a particular manifold generated from the UCS Mechanism [26]. The estimation error for this specie was computed via IV.5 to be 0.0444987. Thus, a tolerance of 0.01 would result in an estimation error that is of the same order as the interpolation error, with a resulting compression rate of 75x. This would need to be repeated for each specie and each manifold, but provides a constructive demonstration of the feasibility of using tensor trains for progress variable tables.

tolerance	compression rate	error, estimation
0.33	149	0.0722304
0.1	149	0.0722304
0.033	149	0.0722304
0.01	75	0.0282988
0.0033	50	0.0166388
0.001	37	0.00775475
0.00033	21	0.00384998
0.0001	9	0.00164248

Table 1 Compression rates and estimation errors for CH_4 .

Moreover, there exists a considerable variability in relative errors among the diverse species encompassed within a manifold. Figure IV.1 illustrates the range of relative errors corresponding to a given compression target. These outcomes are organized into sets, each limited to the number of species in the UCS mechanism (95 species + progress variable source term). Each set comprises species exceeding a compression factor of one while meeting the specified tolerance. As the tolerance is tightened, i.e., reduced, the number of species capable of maintaining the prescribed relative error and achieving compression diminishes. Furthermore, a decrease in tolerance correlates with a reduction in compression. The total compression is also reported using the harmonic mean of the compression of each species and assuming the remaining species are not compressed, ie. their compression is 1.0,

$$\text{total compression} = \left(\frac{n_s}{\sum_{i=1}^{n_c} \frac{1}{C_i} + n_s - n_c} \right)^{-1} \quad (\text{IV.8})$$

where n_c is the number of compressed species in the set and C_i is the compression rate of species i .

It is important to highlight the significance of the manifold. The different species interpolated from the manifold play a vital role in completing the thermodynamic state evaluation and determining specific quantities in the viscous flux, especially the sensible enthalpy transport term. As a result, the accuracy required for minor species that contribute minimally to these thermodynamic properties may not need to be as stringent as for major species. This differs, however, from kinetic model reduction techniques, where it is necessary to resolve minor species or their surrogates, as their branch reactions influence the overall chemical state development, thereby influencing the production and consumption of major species. There exists a plausible spectrum of relative error requirements for different species. This is likely achieved by compressing the minor species, resulting in varied compression levels and a potentially appealing total compression.

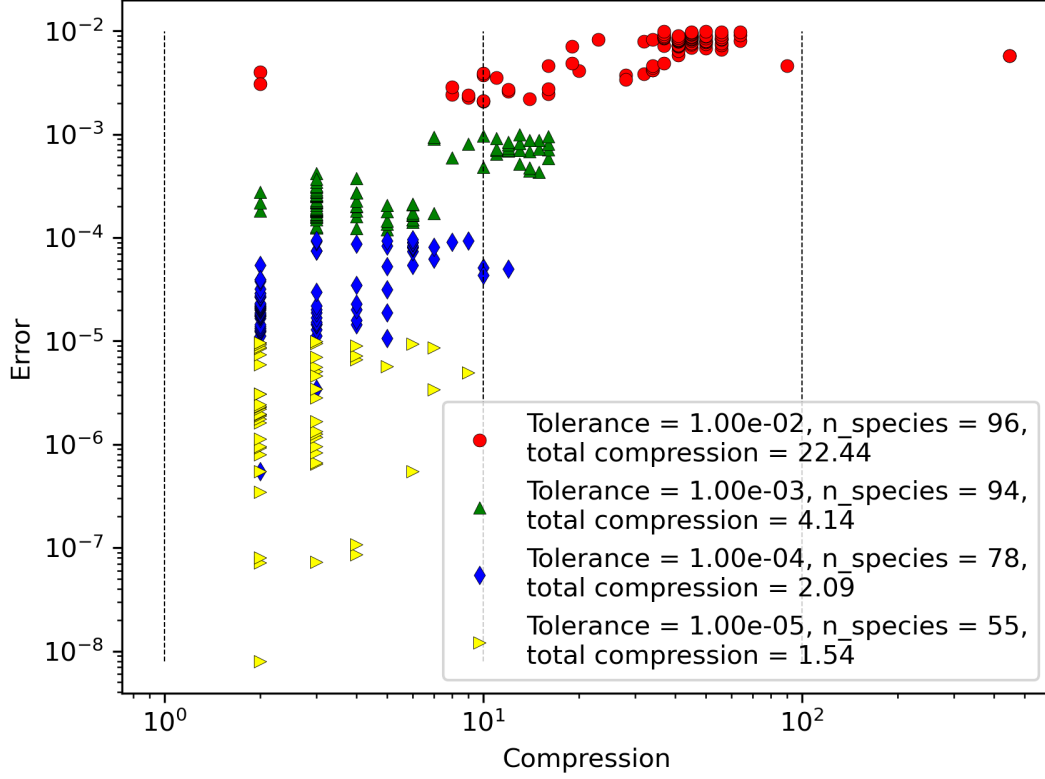


Fig. IV.1 Error associated with each individual species given a desired compression. The colors represent different tolerances for acceptable error.

V. Initial Results, Remarks, and Future Outlook

A. Test Problem Configuration

Here we present a two-dimensional backward with extensional to three-dimensional facing step where fuel is injected along a lower boundary to simulate a solid fuel interface. We assume that the solid fuel is giving off pure methane, CH_4 , and associate it with $Z = 1$ for the mixture fraction. Notably the case is just for demonstration purposes, as a solid fuel, such as Hydroxyl-terminated polybutadiene, would yield larger hydrocarbons in the gas phase, such as butadiene. However, we wished to demonstrate the use of tensor trains first on leaner mechanisms but still utilizing the rich kinetics present in hydrocarbon.

The two dimensional configuration is show in V.1 with appropriate length metrics. The inflow condition is specified using a subsonic characteristic boundary condition utilizing the non-reflective approaches presented in [33],

$$\begin{aligned}
 (v_1, v_2) &= (34, 0) \text{ m/s}, \\
 C_{pv} &= 0.0, \\
 Z &= 0.0, \\
 P &= 101325.0 \text{ Pa}, \\
 T &= 700 \text{ K}.
 \end{aligned} \tag{V.1}$$

The fuel blowing condition is

$$\begin{aligned}
 (v_1, v_2) &= (0, 1.092) \text{ m/s}, \\
 C_{pv} &= 0.0, \\
 Z &= 1.0, \\
 P &= 101325.0 \text{ Pa}, \\
 T &= 700 \text{ K},
 \end{aligned} \tag{V.2}$$

At the bottom boundary, a condition is imposed that represents gas vaporizing and ejecting from the solid fuel surface. The remaining walls are adiabatic, except for a small section near the outflow to assist in outflow condition robustness. The top boundary condition is a slip wall.

The outflow boundary is pressure-specified with a value of $P = 101325.0$ Pa. This does cause initial pressure reflections at the beginning of the simulation. However, as time progresses, the pressure wave leaves through the characteristic inflow. We have found that this is a more robust approach than specifying a pure non-reflective outflow condition, as recirculation zones exiting the domain at start-up may cause issues due to negative velocities for a method that requires outgoing flux.

An unstructured mesh consisting of triangles is constructed using Gmsh [43] Figure V.2 shows the mesh refined to $h = 750 \mu\text{m}$. We utilize h-refinement for all cells, examining a factor of two and four refinement, arriving at a smallest resolution of $h = 187.5 \mu\text{m}$. All simulations are run using cells with $p=2$ and a second-order explicit Runge-Kutta (RK) scheme. We did test additional stages for RK3 and RK4, but no clear advantage in the simulations was observed. The simulation is run until the mixture fraction is well-mixed within the domain. We then use an ignition kernel that specifies $C_{pv} \neq 0$ to initiate combustion. To achieve this, we iterate through all cells and identify degrees of freedom that satisfy the conditions $0.005 < Z < 0.2$, $x_2 > 0.0033$ m, $x_1 < 0.05$ m. The simulation is then run for three flow-through times.

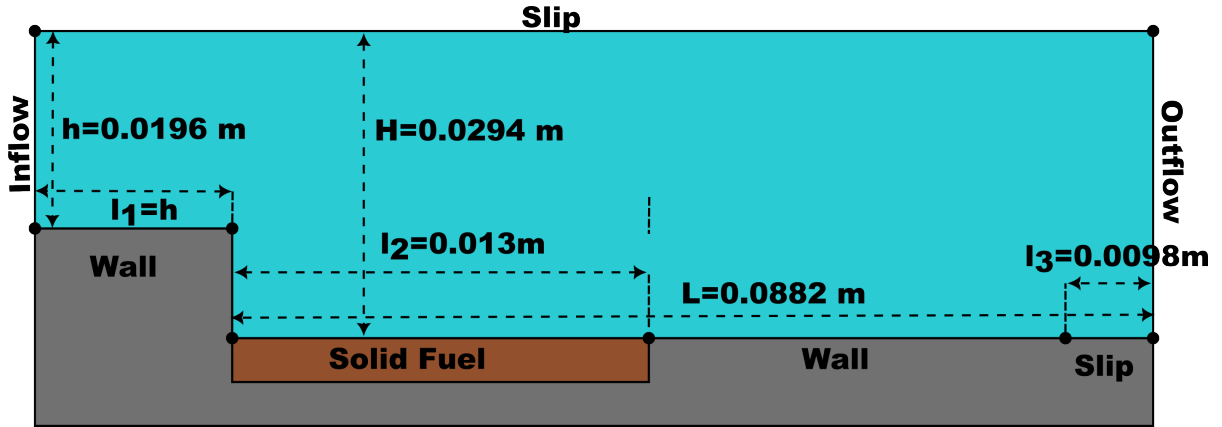


Fig. V.1 Simulated domain of backward facing step with fuel slab.

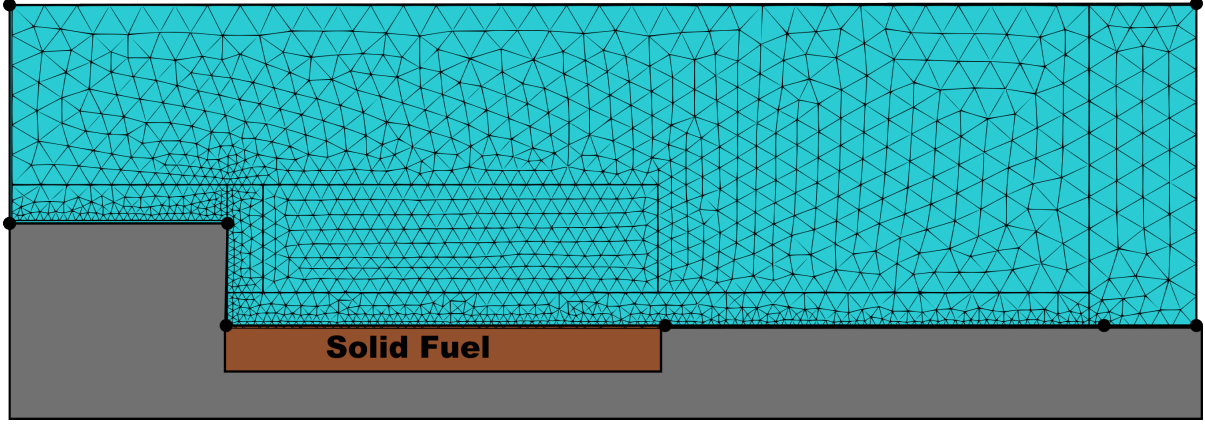


Fig. V.2 Coarse mesh demonstrating clusters of cells for near wall resolution.

B. Two Dimensional Results

In the two-dimensional case, simulations were conducted at target mesh sizes of $h = 750\mu\text{ m}$, $h/2$, and $h/4$. For brevity, we demonstrate the stability of the $h/2$ case with the overlaid mesh in Figures V.3-V.5.

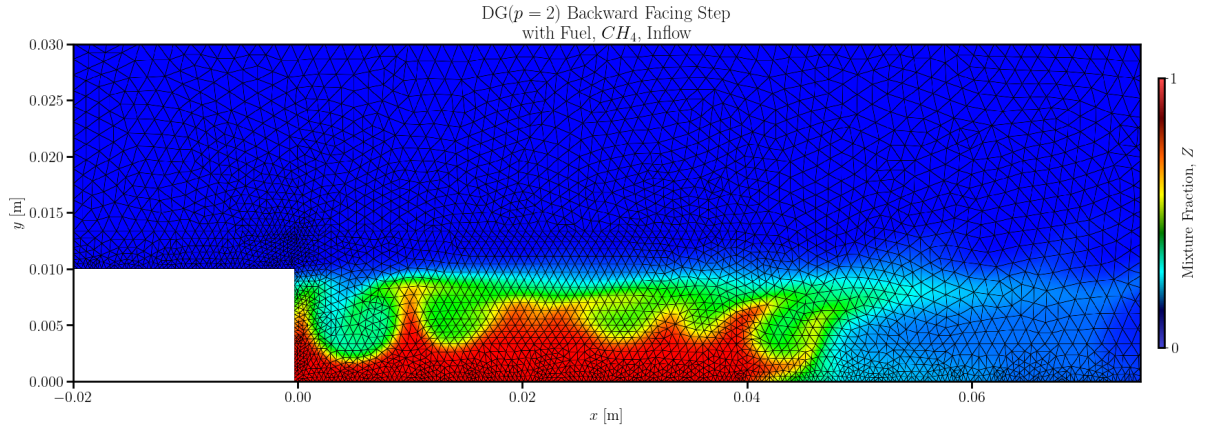


Fig. V.3 $DG(p = 2)$ mixture fraction, Z , solution on $h/2$ mesh. Case was run with FPV manifold and with tensor train compression of manifold.

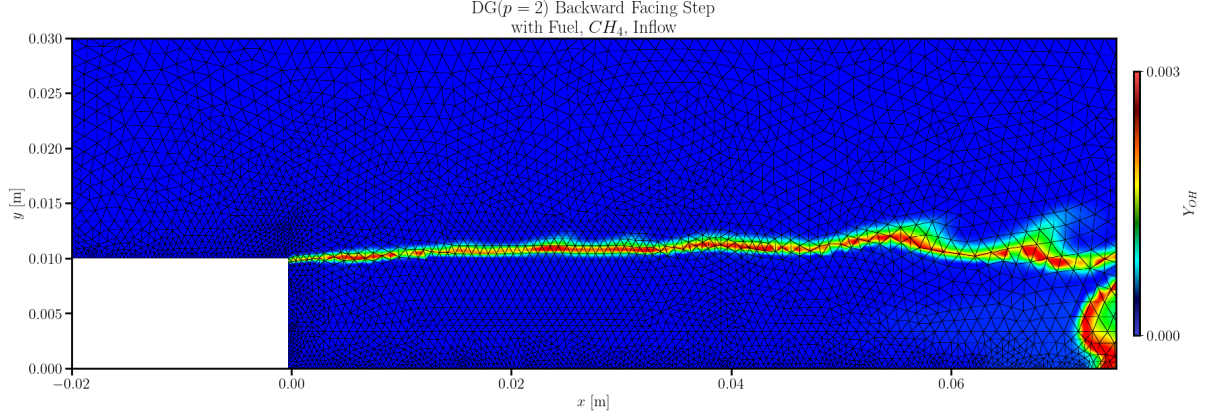


Fig. V.4 $DG(p = 2)$ Hydroxyl, Y_{OH} , solution on $h/2$ mesh. Case was run with FPV manifold and with tensor train compression of manifold.

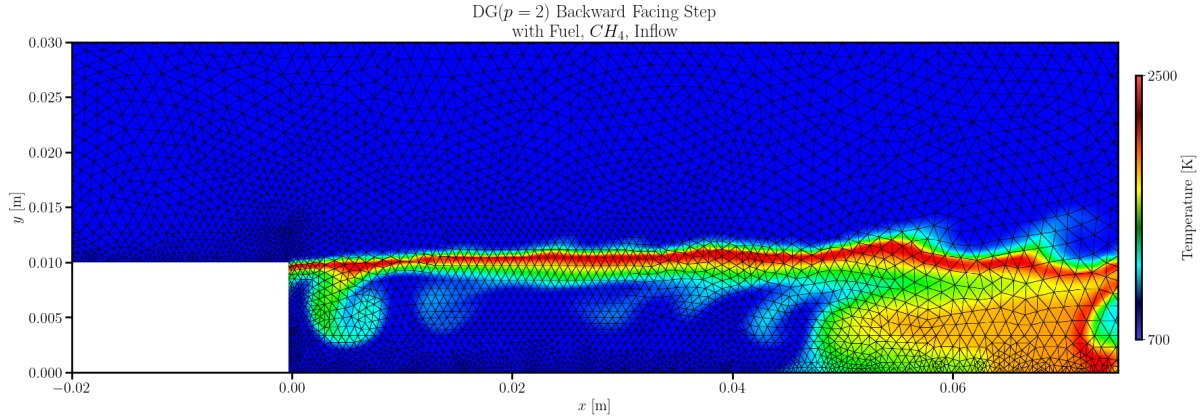


Fig. V.5 $DG(p = 2)$ Temperature solution on $h/2$ mesh. Case was run with FPV manifold and with tensor train compression of manifold.

C. Three Dimensional Results

In the three-dimensional scenario, we broadened the domain to a width equivalent to one step height, incorporating a transverse dimension denoted as x_3 with a specified value of $h = 0.0196$ m (refer to Figure V.1 for the two-dimensional specifications). To enhance stability, we made adjustments by shortening the fuel injection to $l_2/2 = 0.0075$ m and reducing the length of the no-slip wall downstream of the fuel injection. This modification created additional space for the exit of slip walls from the domain.

In the three-dimensional simulation, instability issues arose with vortices exiting the domain. To address this, we extended the length of slip walls reaching the outflow boundary, incorporated larger artificial viscosity, and added temperature-based limiting near the outflow. These changes resulted in a slightly altered target geometry for the three-dimensional results compared to the two-dimensional ones. We plan to update the two-dimensional results to align with these adjustments at the time of the AIAA SciTech conference.

To achieve the three-dimensional extension, we employed the extrusion function in Gmsh, adding equally spaced cells in the x_3 direction with a target resolution of $500\mu\text{m}$. The cells in the x_1 - x_2 plane maintained the same target mesh sizes. Notably, only tetrahedral cells were generated. In this expanded domain, the inflow quantities now include a requisite third dimension for all velocities, denoted as v_3 . We specified no flux in the x_3 direction, setting $v_3 = 0$ m/s.

An initial solution is generated with a $DG(p = 0)$ approach. Once the mixing and under-resolved velocity field have been developed, we proceed to run the simulation using $DG(p = 1)$ and subsequently $DG(p = 2)$ as the final polynomial

degrees. After achieving a well-developed solution with $DG(p = 2)$, we then implement a global h-refinement. Once this refinement is fully developed, we iterate use the same ignition procedure as the two-dimensional case.

Figures V.6-V.10 display an instantaneous temperature solution of the three-dimensional results on the $h/2$ mesh. The three-dimensional results have third-dimensional dependency, highlighting the need for the simulations to scale well. A flame is attached to the corner of the backward facing step with a recirculation zone stabilizing the flame behind the fuel injection. High temperature regions are also found along the wall downstream of the fuel. A thin reaction layer is found along the no-slip adiabatic wall downstream of the fuel injection.

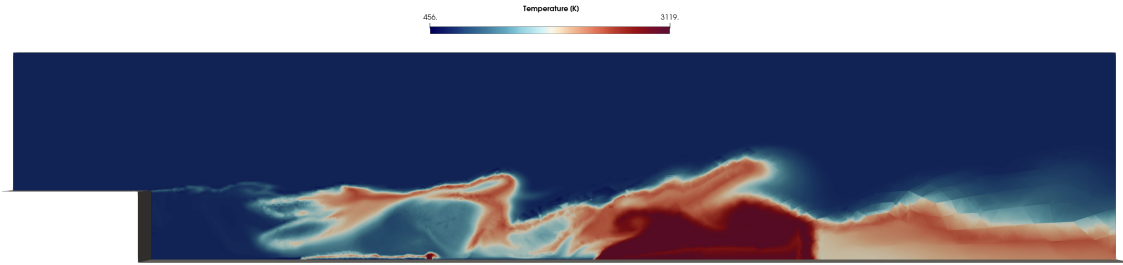


Fig. V.6 $DG(p = 2)$ Temperature solution on $h/2$ mesh plane with $x_3 = 0$ m.

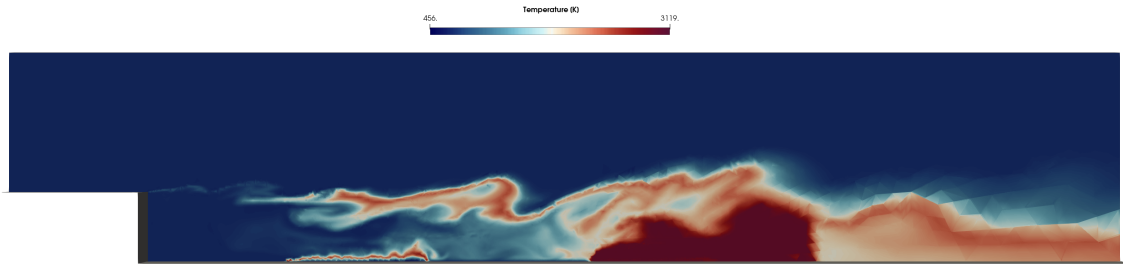


Fig. V.7 $DG(p = 2)$ Temperature solution on $h/2$ mesh plane with $x_3 = 0.0025$ m.

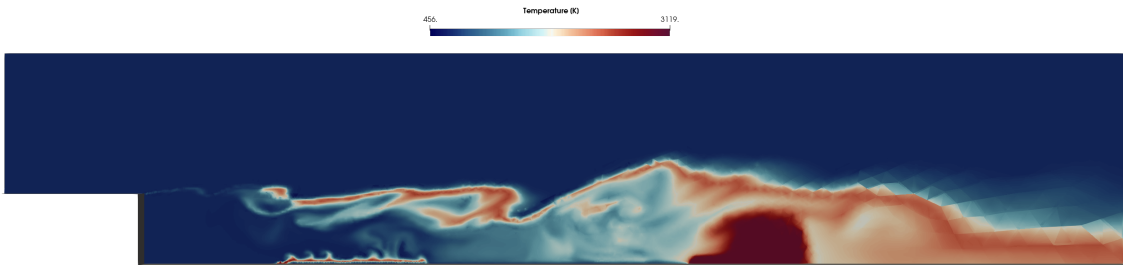


Fig. V.8 Temperature solution on $h/2$ mesh plane with $x_3 = 0.005$ m.

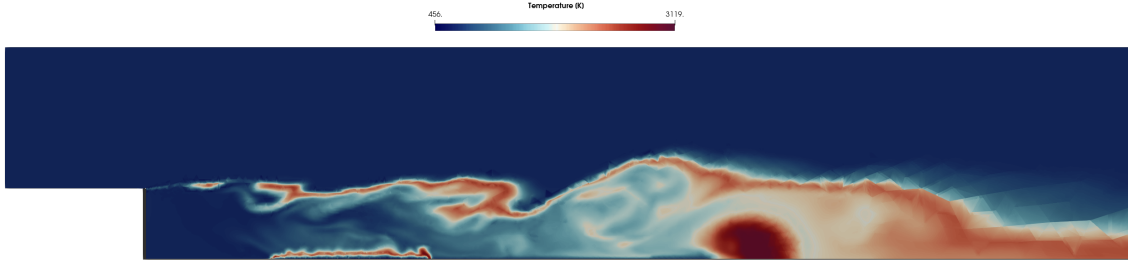


Fig. V.9 $DG(p = 2)$ Temperature solution on $h/2$ mesh plane with $x_3 = 0.0075$ m.

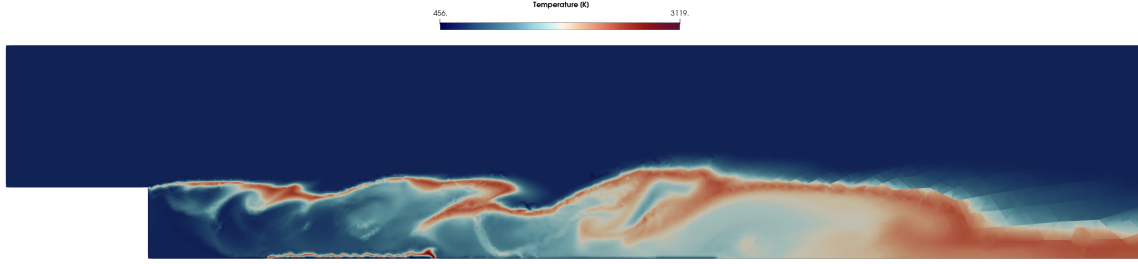


Fig. V.10 $DG(p = 2)$ Temperature solution on $h/2$ mesh plane with $x_3 = 0.0095$ m.

We conducted additional evaluations employing tensor trains for the aforementioned cases, and our observations revealed no discernible qualitative distinctions. Our forthcoming analysis will focus on presenting quantitative metrics to comprehensively elucidate the comparative outcomes.

At the time of writing this manuscript we are in the process of acquiring the following information that will be present at the AIAA 2024 SciTech:

- 1) Compression for even larger manifolds. We will concentrate on the largest number of species that we have available for demonstration purposes.
- 2) Qualitative stability of using tensor trains in place of the interpolation of the flamelet manifold.
- 3) Behavior of system response quantities (SRQ) over a simulation time. We will start the three-dimensional simulations from the same initial condition and measure the divergence of the SRQs as the simulation is performed and if the departure using the tensor train is significant.
- 4) We will note any special performance increases. As of now, the three dimensional simulations are run on 8 a100 NVIDIA GPUs. We will report an qualitative increase in simulation efficiency.

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